

INTRODUCTION TO

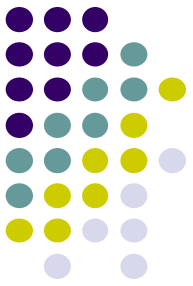
Morphological Image Processing

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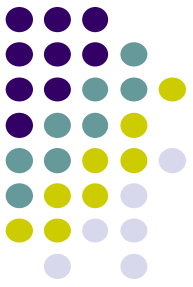
Faculty of Computer Engineering

University of Isfahan



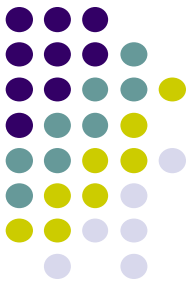
Introduction

- The word *morphology* commonly denotes a branch of biology that deals with the form and structure of animals and plants.
- Therefore, morphological operations are intended to affect the *shape of the object*



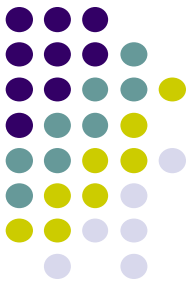
We use mathematical morphology as a tool for **extracting image components** that are useful in the representation and description of region shape, such as:

- boundaries extraction
- skeletons
- convex hull
- morphological filtering
- thinning
- pruning



Sets in mathematic morphology represent objects in an image:

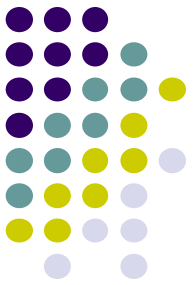
- binary image (0 = black, 1 = white) :
the element of the set is the coordinates (x,y) of pixel belong to the object Z^2
- gray-scaled image :
the element of the set is the coordinates (x,y) of pixel belong to the object and the gray levels Z^3



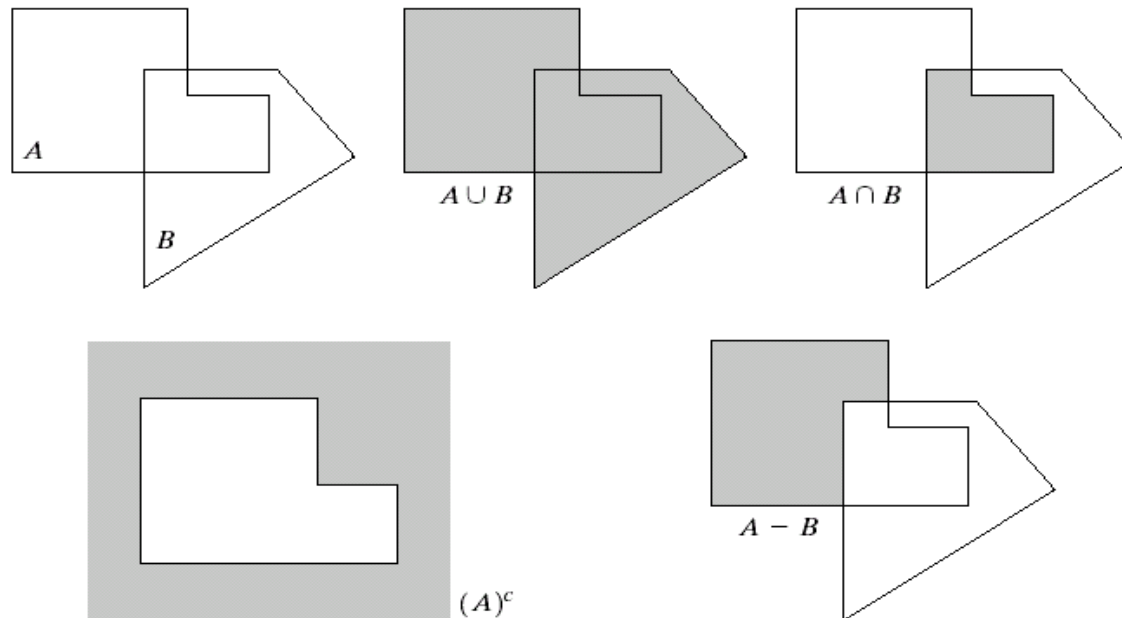
Notations:

- A is a set in Z^2 ,
- $a=(a_1,a_2)$ an element of A , $a \in A$ If not, then $a \notin A$
- $\emptyset$: null (empty) set
- A subset of B : $A \subseteq B$
- Union of A and B : $C=A \cup B$
- Intersection of A and B : $D=A \cap B$
- Disjoint sets: $A \cap B = \emptyset$
- Complement of A : $A^c = \{x | x \notin A\}$
- Difference of A and B :

$$A-B = \{x | x \in A, x \notin B\} = A \cap B^c$$



Operators by examples:



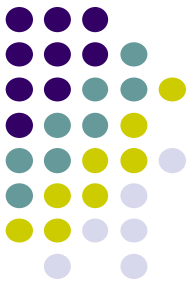
a	b	c
d	e	

FIGURE 9.1

(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .



Reflection and Translation by examples:



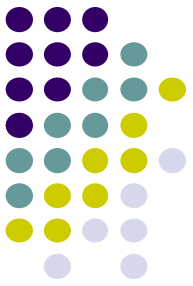
Need for a reference point.

- **Reflection** of B:

$$\hat{B} = \{x | x = -b, \text{ for } b \in B\}$$

- **Translation** of A by $x = (x_1, x_2)$, denoted by $(A)_x$ is defined as:

$$(A)_x = \{c | c = a + x, \text{ for } a \in A\}$$

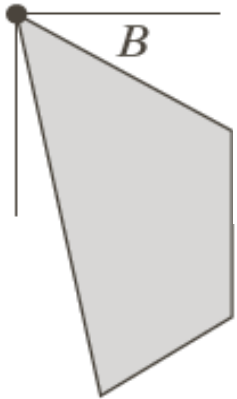


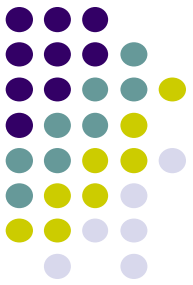
Example: Reflection and Translation

a b c

FIGURE 9.1

(a) A set, (b) its reflection, and (c) its translation by z .





Basic Logic Operations:

TABLE 9.1

The three basic logical operations.

p	q	p AND q (also $p \cdot q$)	p OR q (also $p + q$)	NOT (p) (also $\bar{p}$)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

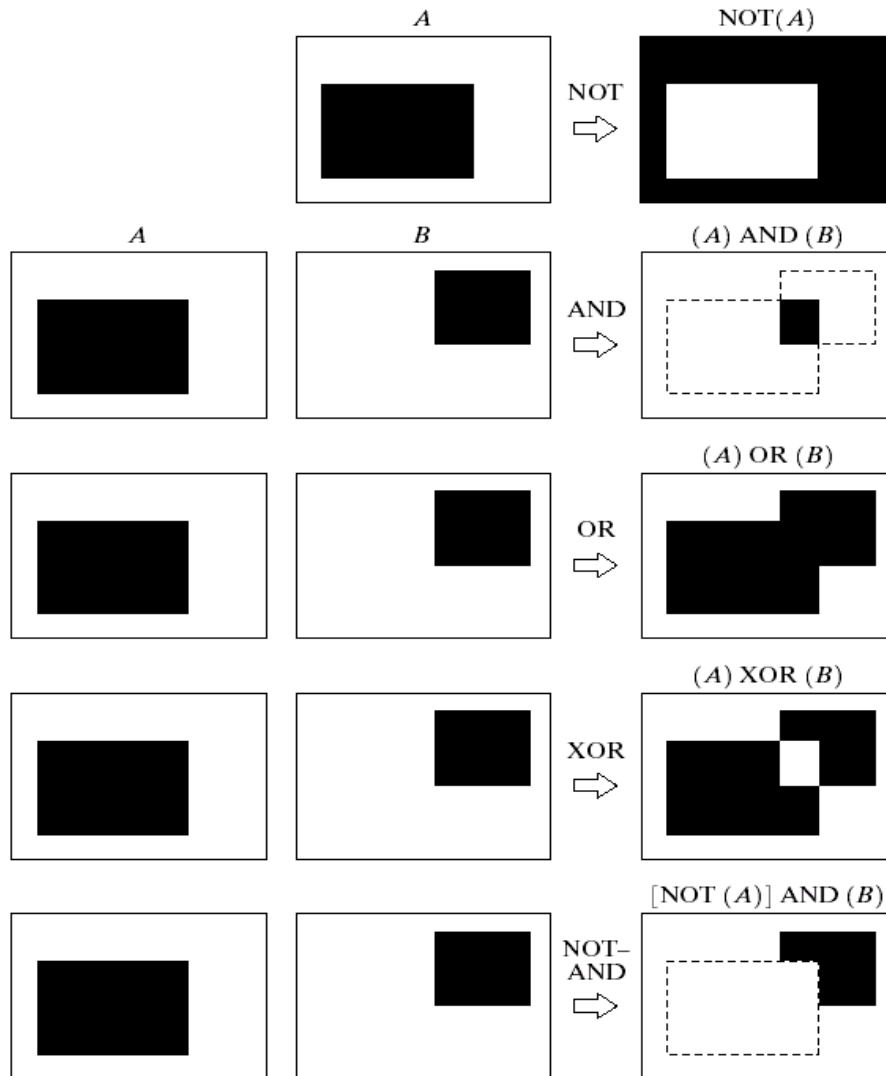
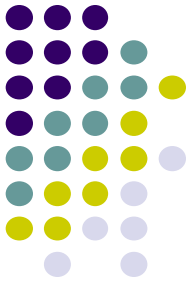
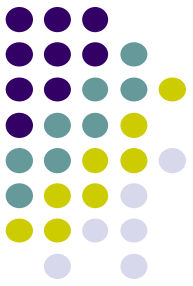


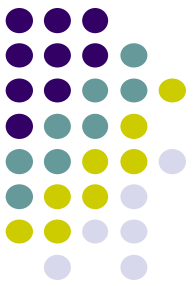
FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.





Structure elements (SE)

Small sets or sub-images used to probe an image under study for properties of interest



Examples: Structuring Elements (1)

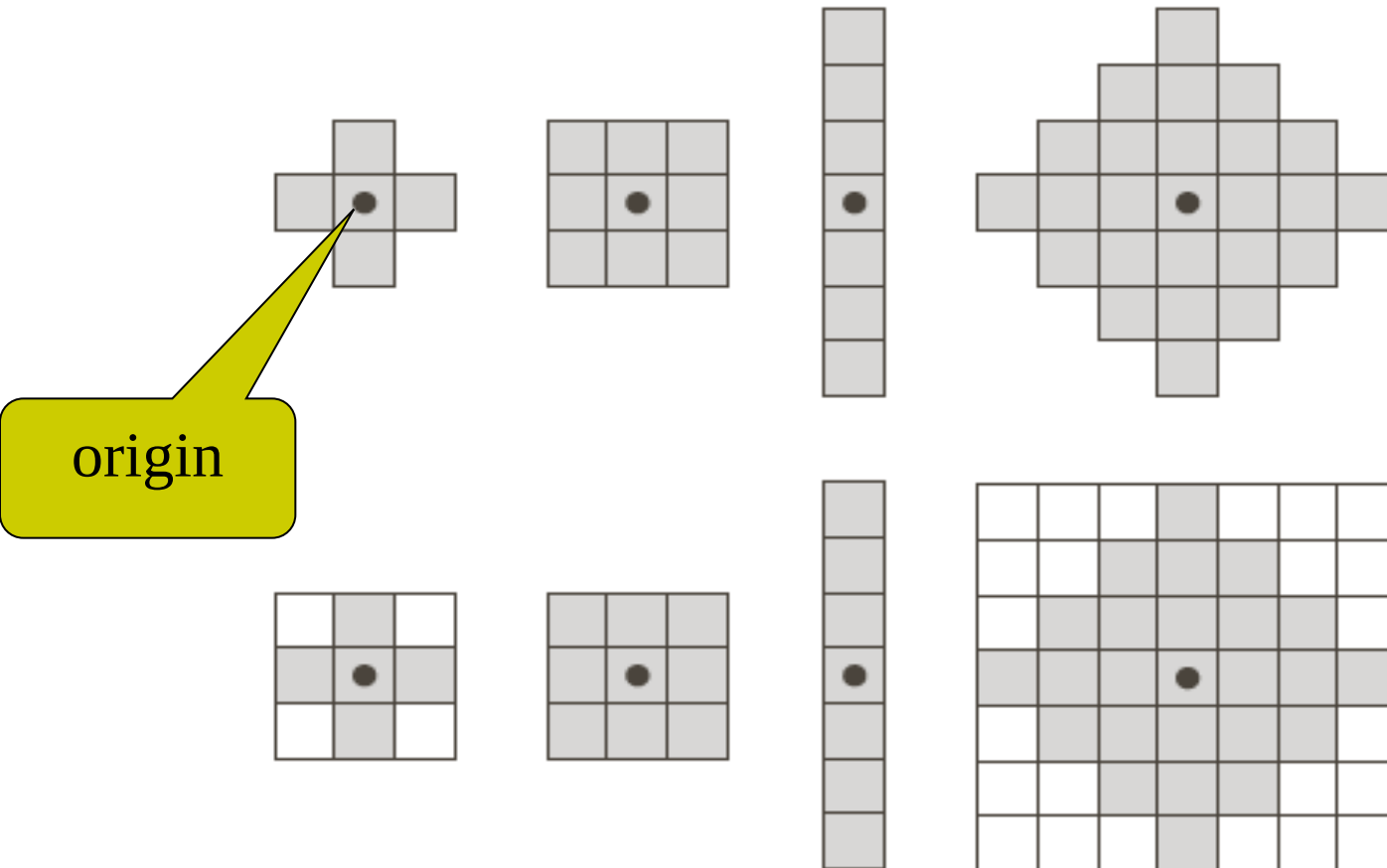
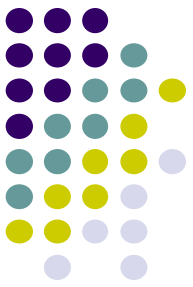
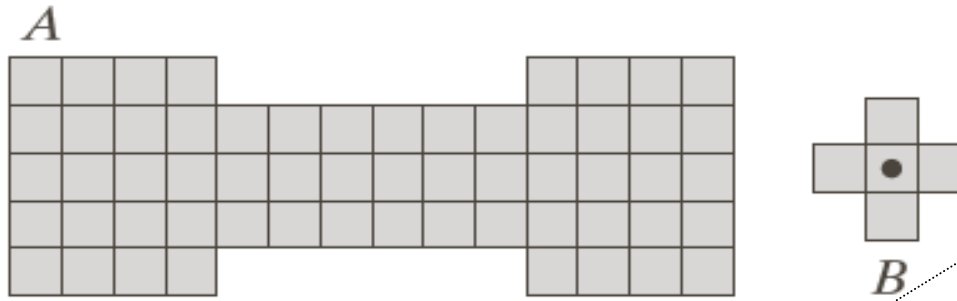


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.



Examples: Structuring Elements (2)

Accommodate the entire structuring elements when its origin is on the border of the



Origin of B visits every element of

A

At each location of the origin of B, if B is completely contained in A, then the location is a member of

the new set, otherwise it is not a member of the new set.

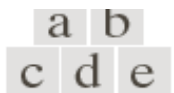
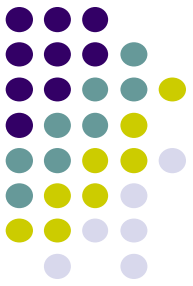


FIGURE 9.3 (a) A set (each shaded square is a member of the set). (b) A structuring element. (c) The set padded with background elements to form a rectangular array and provide a background border. (d) Structuring element as a rectangular array. (e) Set processed by the structuring element.



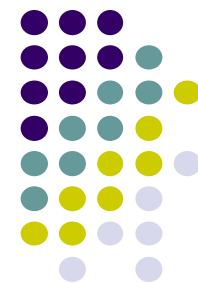
Minkowski Set Operations:

The basic *Minkowski set operations*--addition and subtraction--can now be defined.

First we note that the individual elements that comprise B are not only pixels but also *vectors* as they have a clear coordinate position with respect to $[0,0]$. Given two sets A and B :

Minkowski addition - $A \oplus B = \bigcup_{\beta \in B} (A + \beta)$

Minkowski subtraction - $A \ominus B = \bigcap_{\beta \in B} (A + \beta)$



Basic Operators:

From these two Minkowski operations we define the fundamental mathematical morphology operations *dilation* and *erosion*:

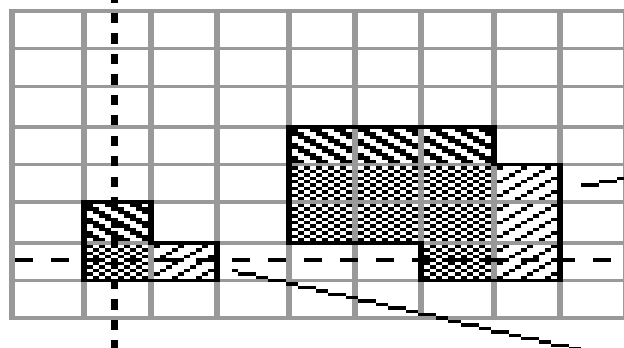
Dilation -

$$D(A, B) = A \oplus B = \bigcup_{\beta \in B} (A + \beta)$$

Erosion -

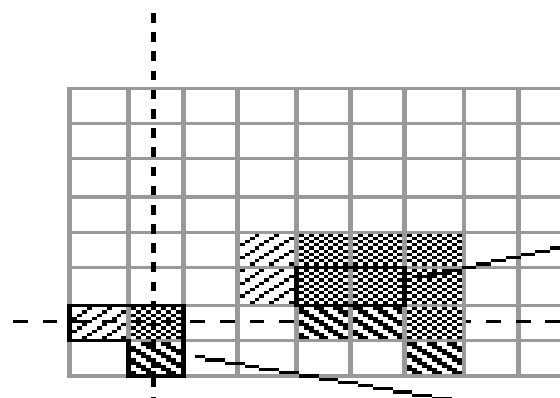
$$E(A, B) = A \ominus (-B) = \bigcap_{\beta \in B} (A - \beta)$$

where $-B = \{-\beta | \beta \in B\}$



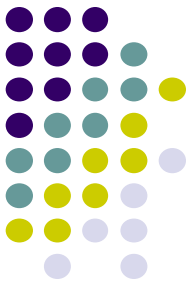
$D(A, B)$

B



$E(A, B)$

$-B$



Dilation

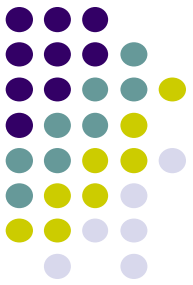
- Dilation Definition by Reflection & Translation

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \emptyset\} = \{z | (\hat{B})_z \cap A \subseteq A\}$$

1. Obtaining the reflection of B about its origin and then shifting this reflection by x
2. The dilation of A by B then is the set of all x displacements such that A and the reflection of B overlap by at least one nonzero element...
(B is the structuring element in dilation)

Relation to Convolution mask:

- Flipping
- Overlapping
- Dilation is typically applied to binary image, but there are versions that work on gray scale image.



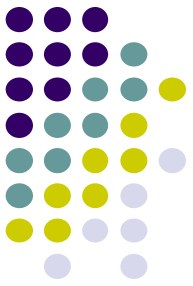
Suppose that X is the set of coordinates corresponding to the input binary image, and that K is the set of coordinates for the structuring element.

Let Kx denote the translation of K so that its origin is at x .

Then the dilation of X by K is simply the set of all points x such that the intersection of Kx with X is non-empty.



Example 1:



Suppose that the structuring element is a 3×3 square .

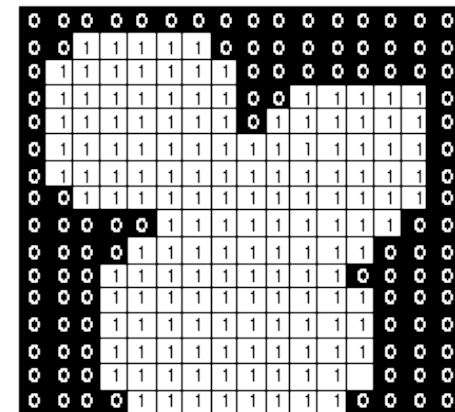
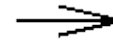
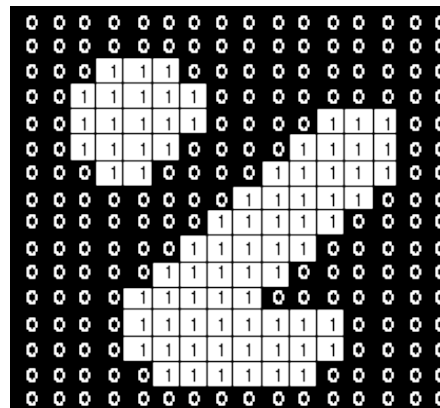
Note that in subsequent diagrams, foreground pixels are represented by 1's and background pixels by 0's.

To compute the dilation of a binary input image by this structuring element, we superimpose the structuring element on top of the input image so that the origin of the structuring element coincides with the input pixel position.

If the center pixel in the structuring element coincides with a foreground pixel in the image underneath, then the input pixel is set to the foreground value.

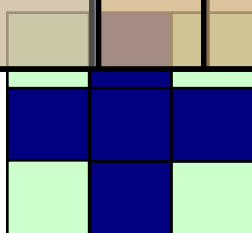
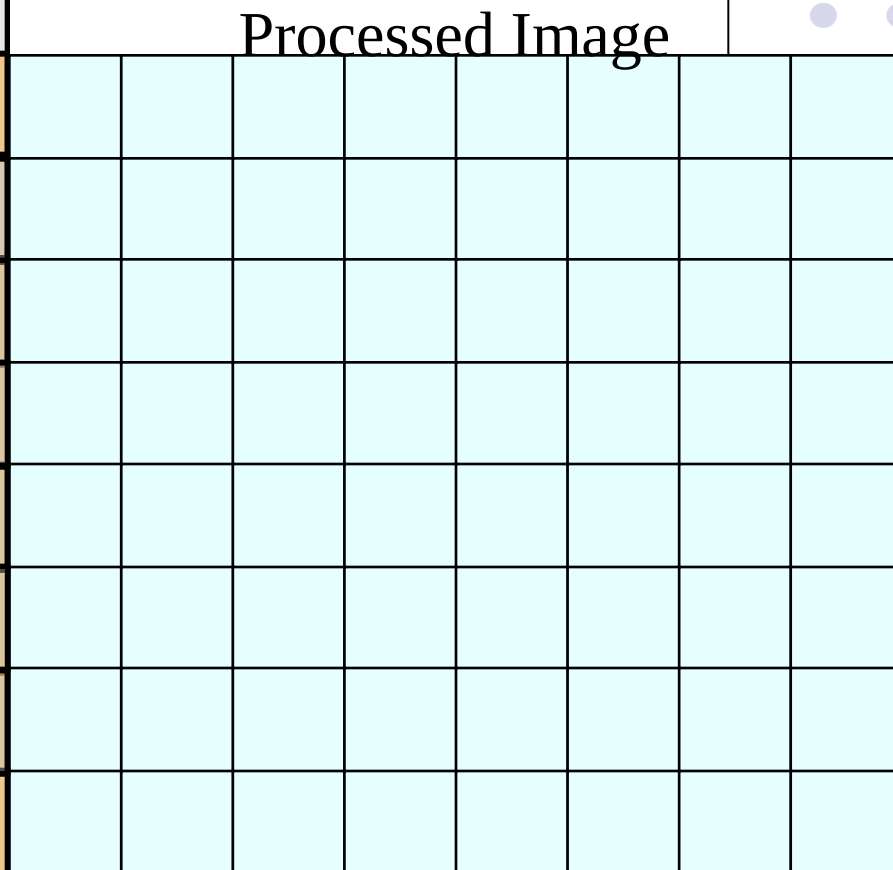
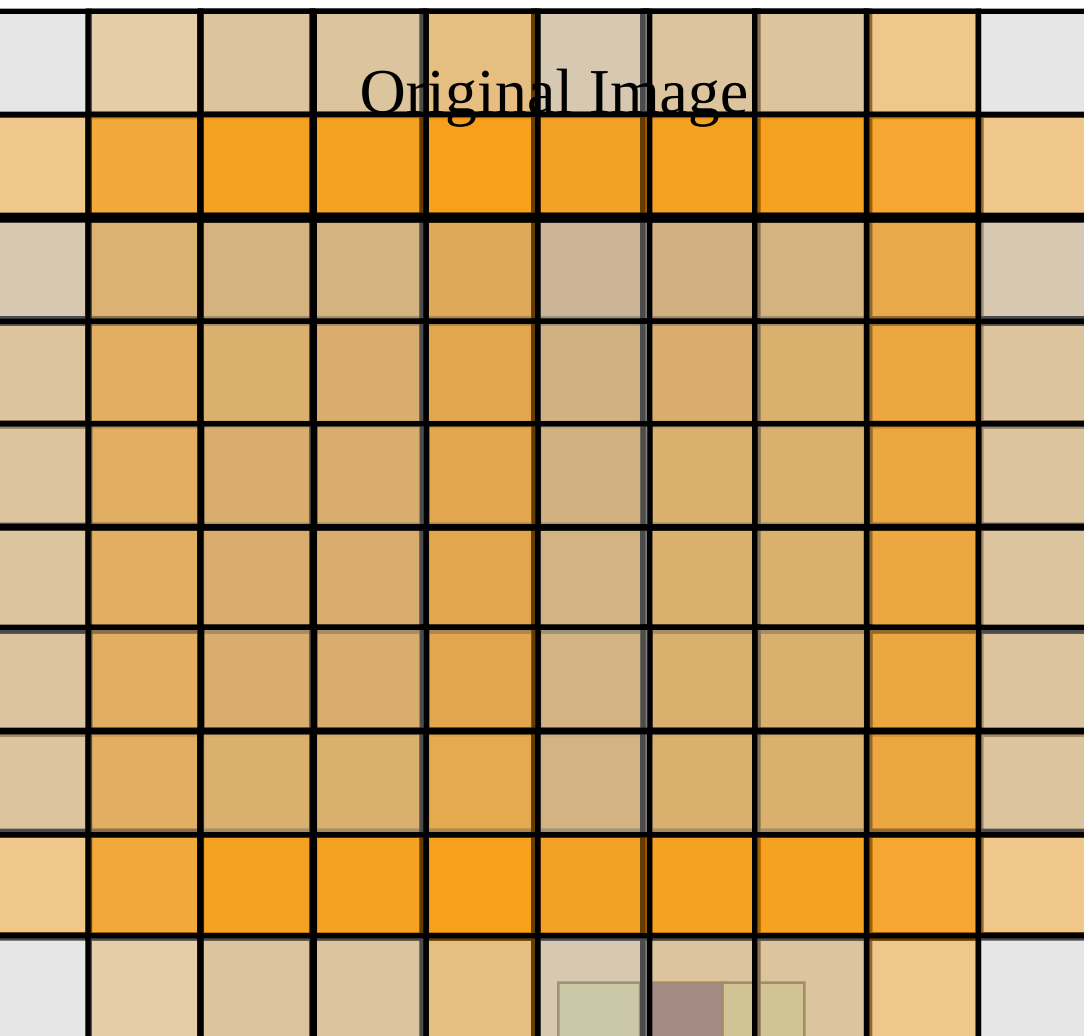
1	1	1
1	1	1
1	1	1

Structuring element





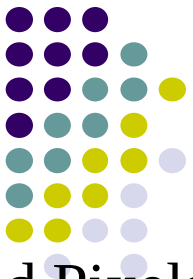
Dilation Example 2



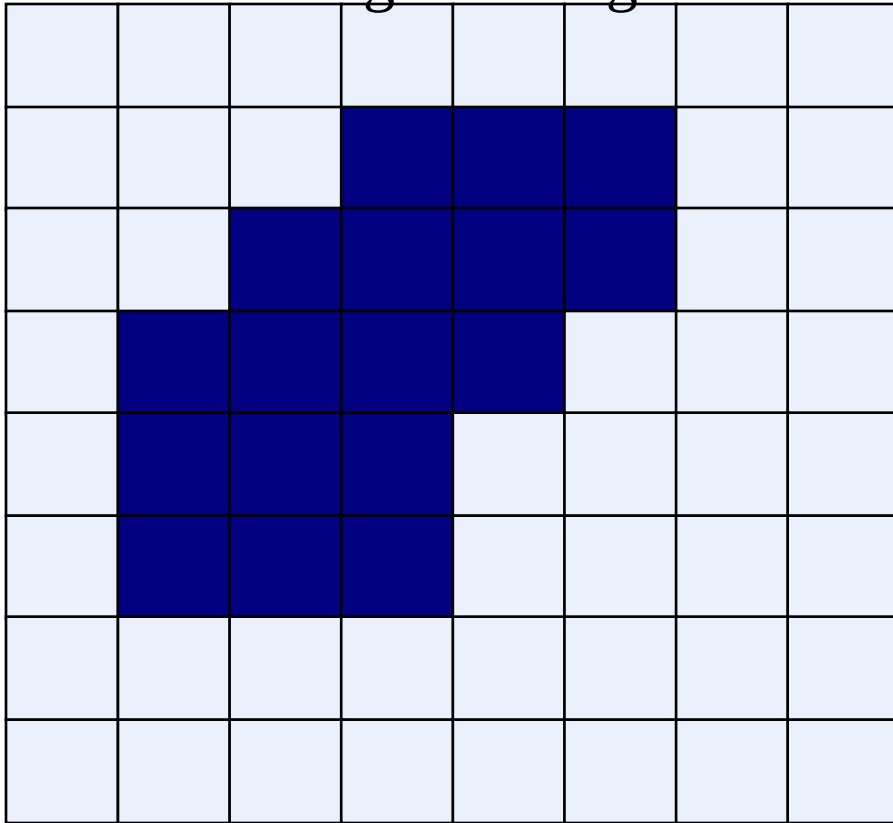
Structuring Element



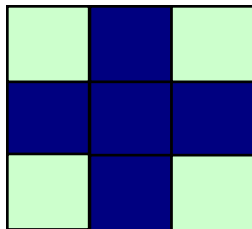
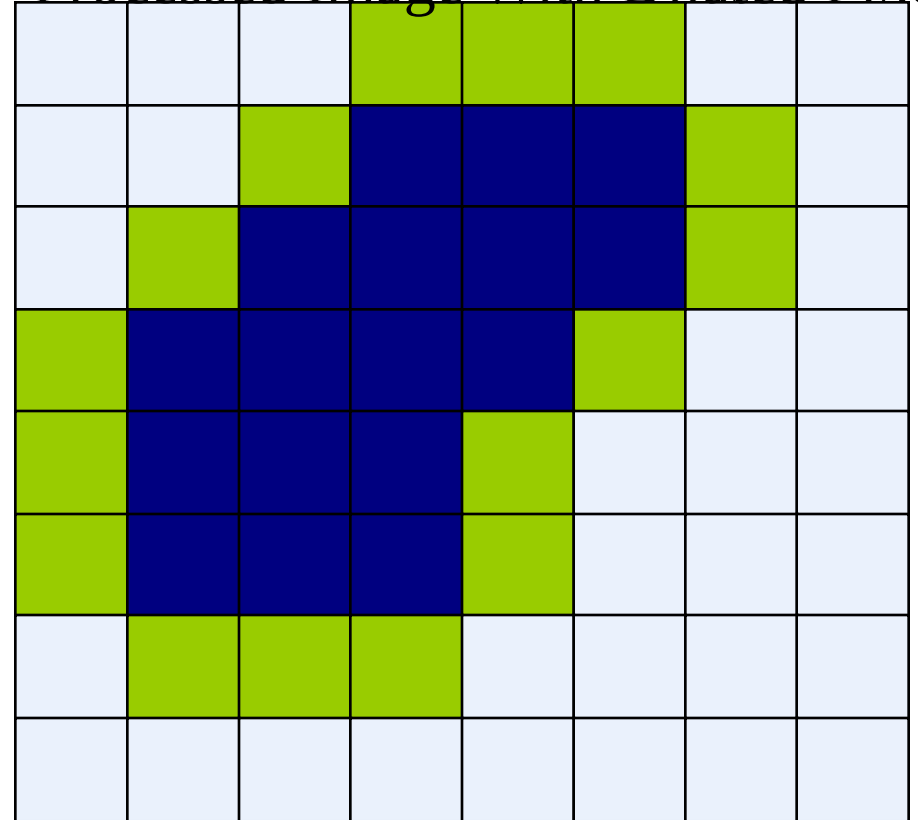
Dilation Example 2 cont.



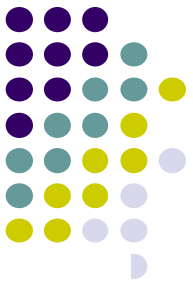
Original Image



Processed Image With Dilated Pixels



Structuring Element



Dilation Example 3



Original image

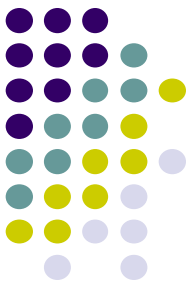
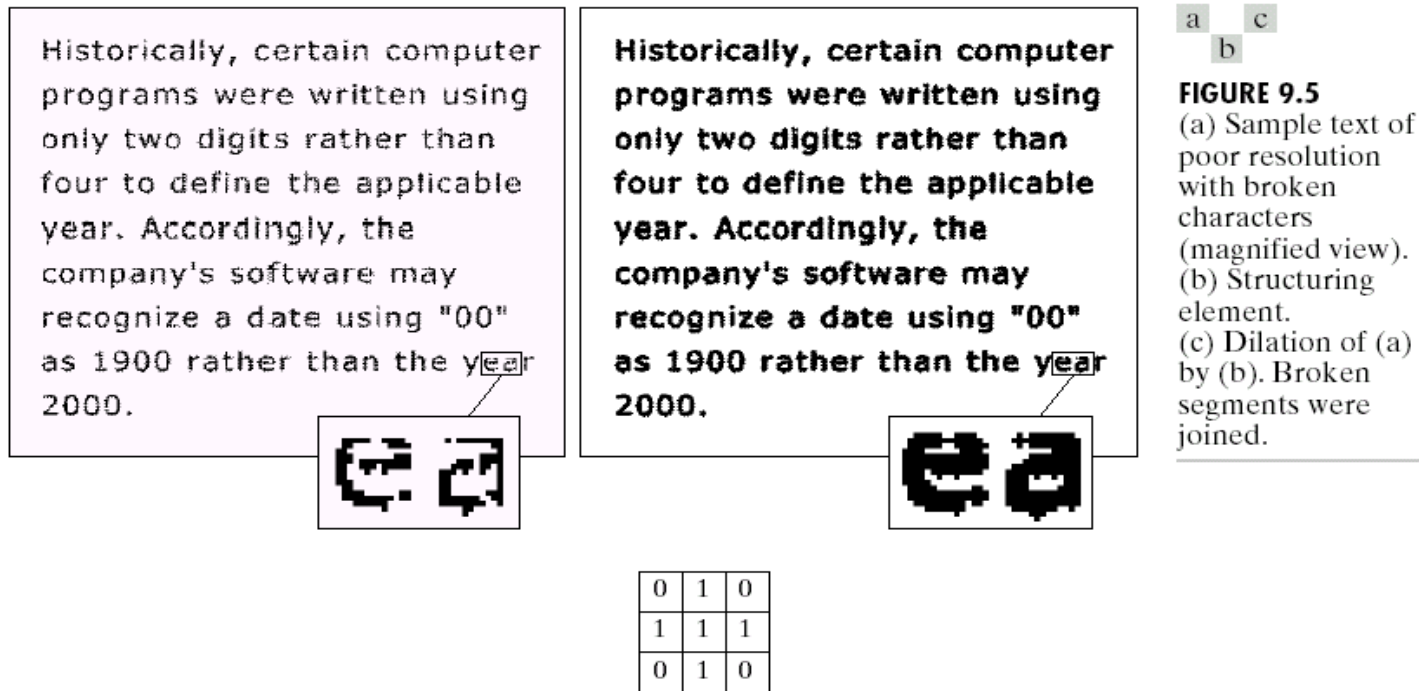


Dilation by 3×3
square structuring
element



Dilation by 5×5
square structuring
element

Watch out: In these examples a 1 refers to a black pixel!



❑ The basic effect of the operator on a binary image is to gradually enlarge the boundaries of regions of foreground pixels (*i.e.* white pixels, typically).

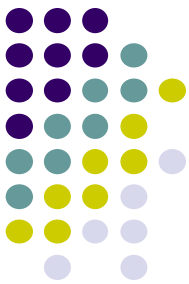
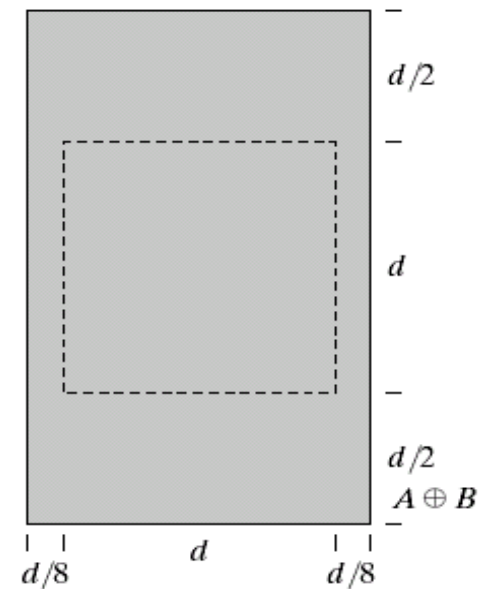
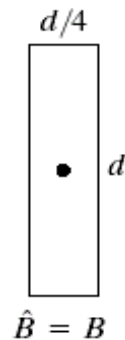
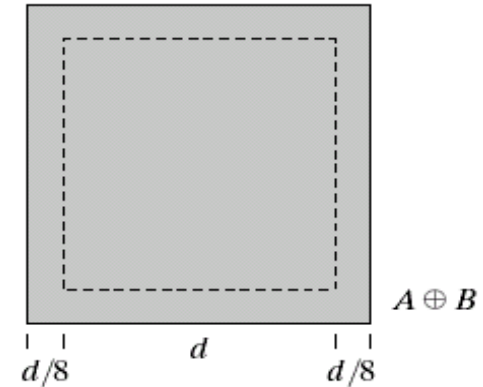
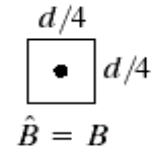
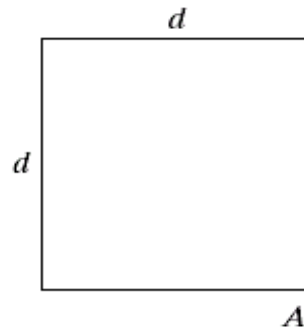


Dilation by example:

a	b	c
d		e

FIGURE 9.4

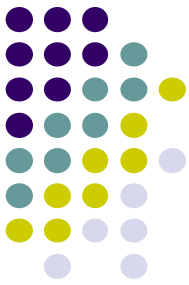
- (a) Set A .
 (b) Square structuring element (dot is the center).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element.

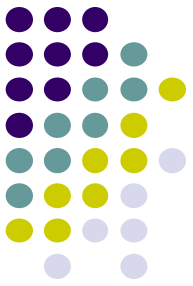




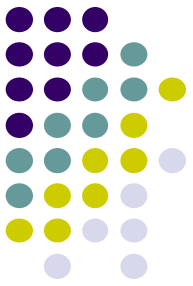
- MATLAB CODE: MorphExample01

```
clear all;
B=zeros(256,256); % create 256x256 image
for i=128:160
    for j=128:160 B(i,j)=1;
    end
end
for i=20:25
    for j=30:190 B(i,j)=1;
    end
end
i=1;
while i<40 % generate 40 random pixels
x=uint8(rand*255); y=uint8(rand*255);
B(x,y)=1;i=i+1;
end
imshow(B); title('Original image');
K3=ones(3); K5=ones(5); K7=ones(7); K9=ones(9); % create kernels
B3=imdilate(B,K3); B5=imdilate(B,K5); % apply dilation
B7=imdilate(B,K7); B9=imdilate(B,K9);
figure; imshow(B3); title('Dilated image with 3 by 3');
figure; imshow(B5); title('Dilated image with 5 by 5');
figure; imshow(B7); title('Dilated image with 7 by 7');
figure; imshow(B9); title('Dilated image with 9 by 9');
```

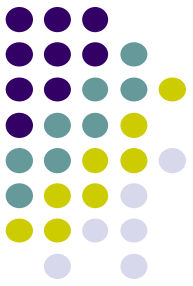




- Dilation is typically applied to binary image, but there are versions that work on gray scale image.
- The basic effect of the operator on a binary image is to gradually enlarge the boundaries of regions of foreground pixels (*i.e.* white pixels, typically).
- Thus areas of foreground pixels grow in size while holes within those regions become smaller.



- **Summary effects of dilation:**
 - Expand/enlarge objects in the image
 - Fill gaps or bays of insufficient width
 - Fill small holes of sufficiently small size
 - Connects objects separated by a distance less than the size of the window



2. Erosion

- **Erosion Definition:**

$$A \ominus B = \{x \mid (B)_x \subseteq A\}$$

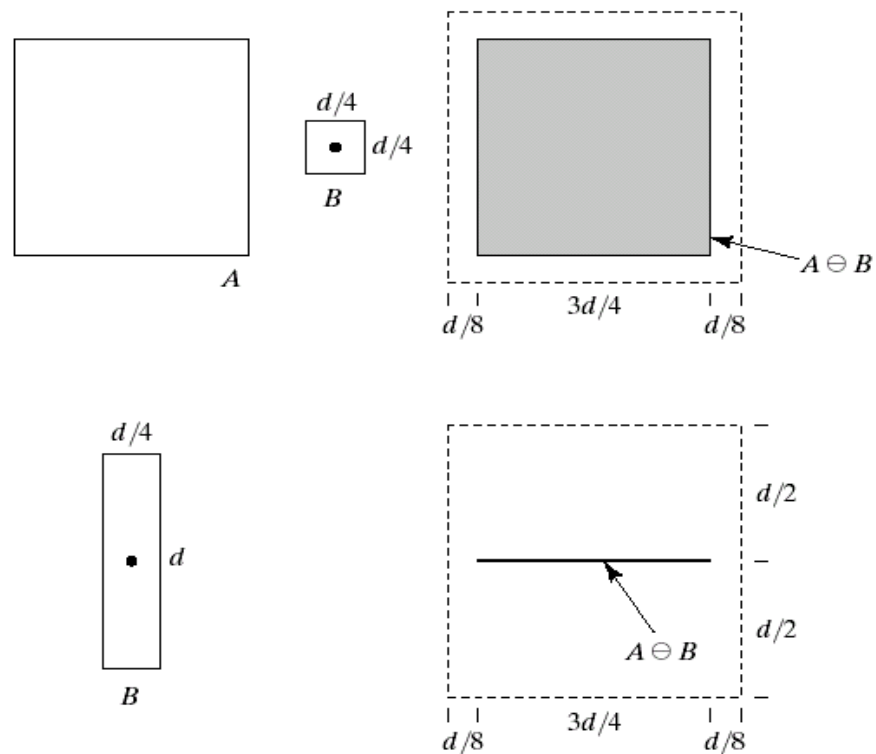
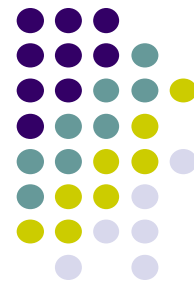
i.e. the erosion of A by B is the set of all points x such that B, translated by x, is contained in A.

It is typically applied to binary image, but there are versions that work on gray scale image.

- The basic effect of the operator on a binary image is to erode away the boundaries of regions of foreground pixels (*i.e.* white pixels, typically).
- Thus areas of foreground pixels shrink in size, and holes within those areas become larger.



$$A \ominus B = \{x \mid (B)_x \subseteq A\}$$

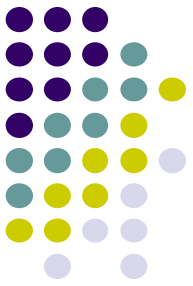


a	b	c
d		e

FIGURE 9.6 (a) Set A . (b) Square structuring element. (c) Erosion of A by B , shown shaded. (d) Elongated structuring element. (e) Erosion of A using this element.



Erosion Example 1:



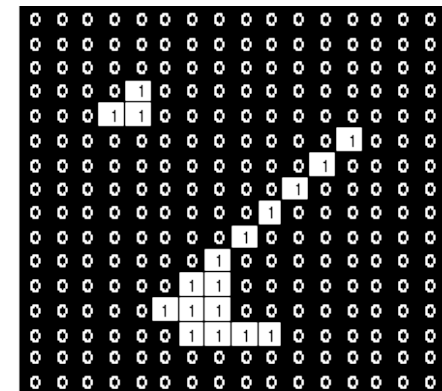
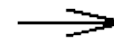
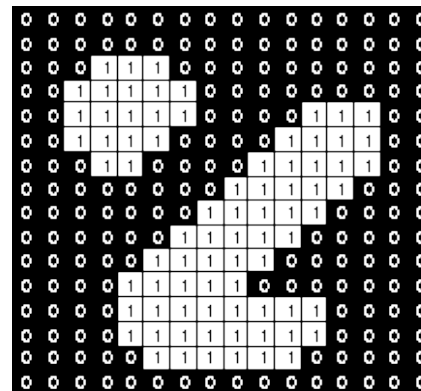
Suppose that the structuring element is a 3×3 square .

Note that in subsequent diagrams, foreground pixels are represented by 1's and background pixels by 0's.

The structuring element is now superimposed over each foreground pixel (input pixel) in the image. If all the pixels below the structuring element are foreground pixels then the input pixel retains it's value. But if any of the pixels is a background pixel then the input pixel gets the background pixel value.

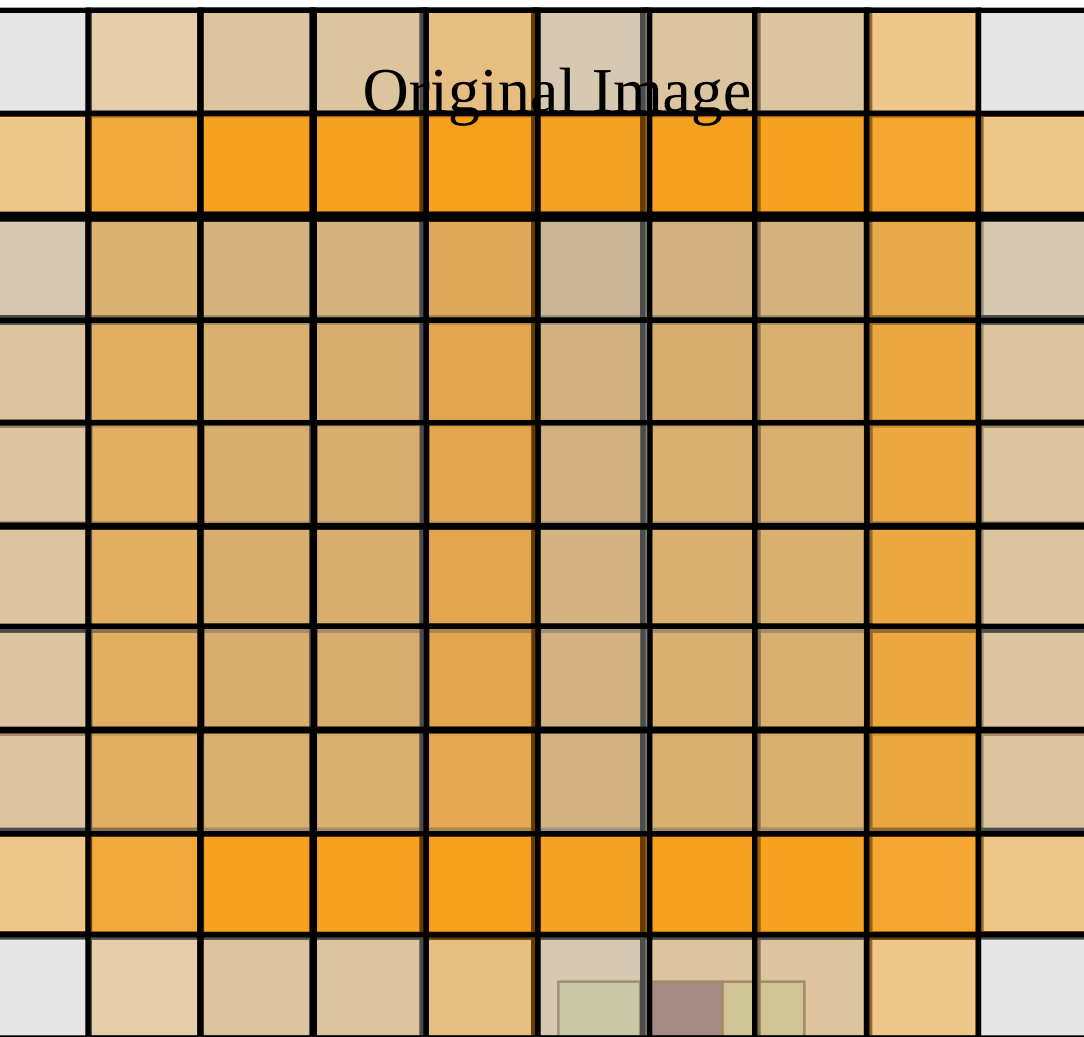
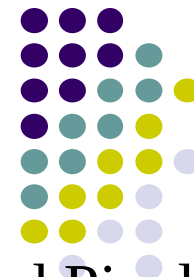
1	1	1
1	1	1
1	1	1

Structuring element

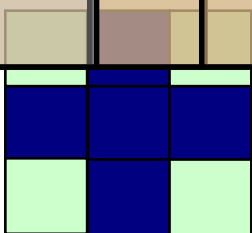
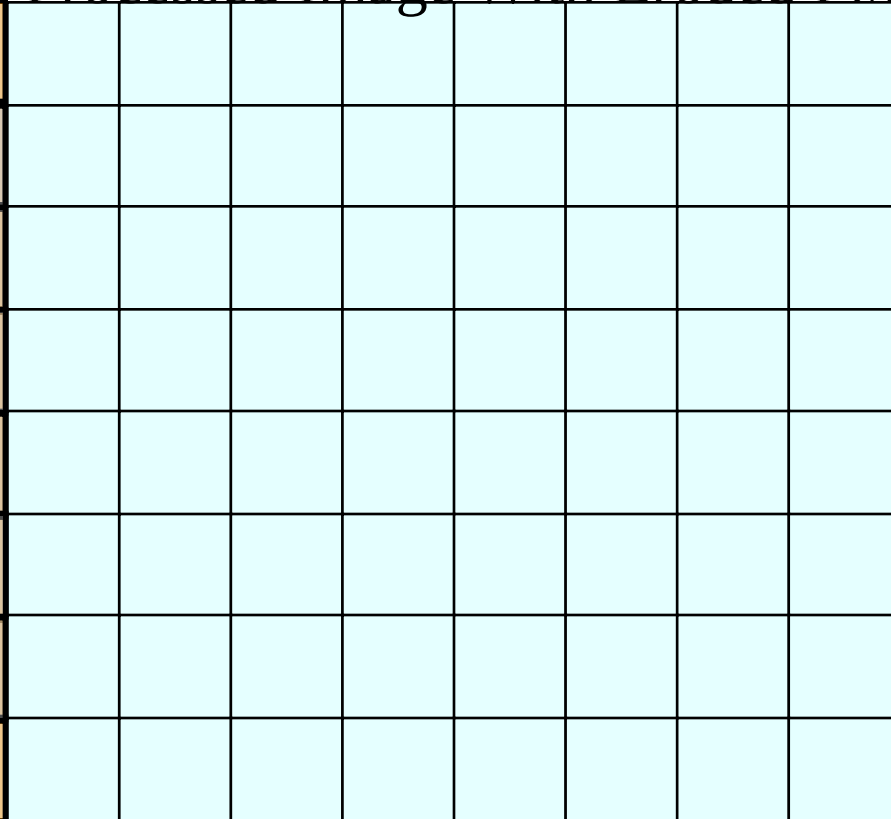




Erosion Example 2



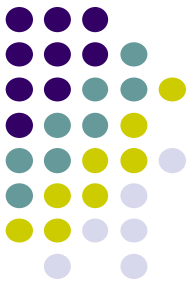
Processed Image With Eroded Pixels



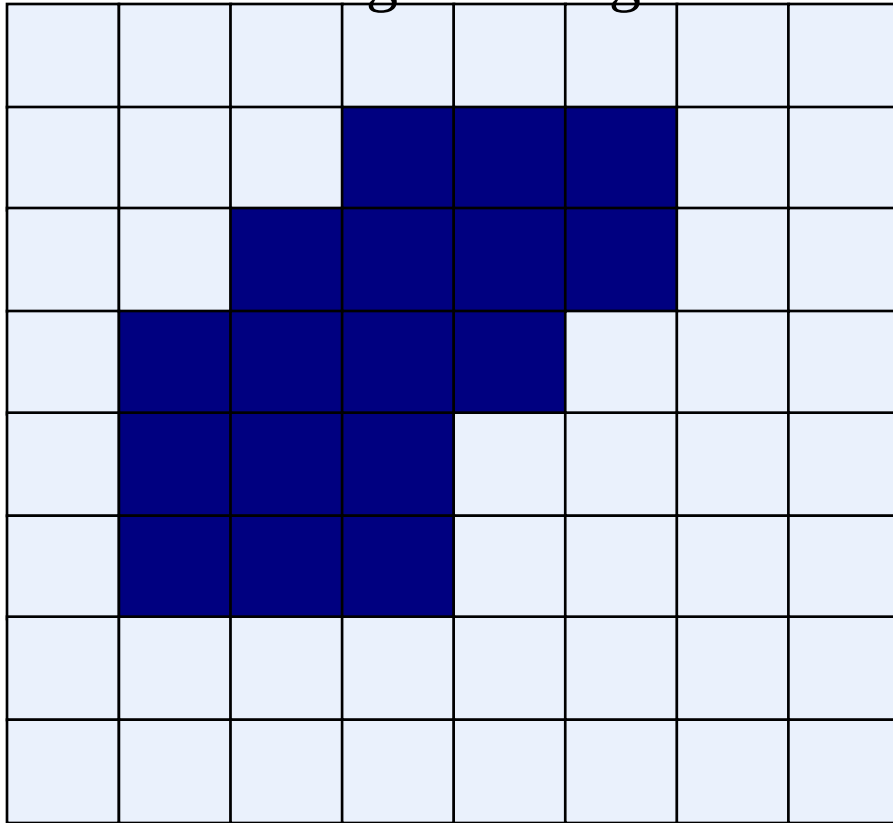
Structuring Element



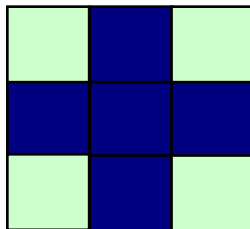
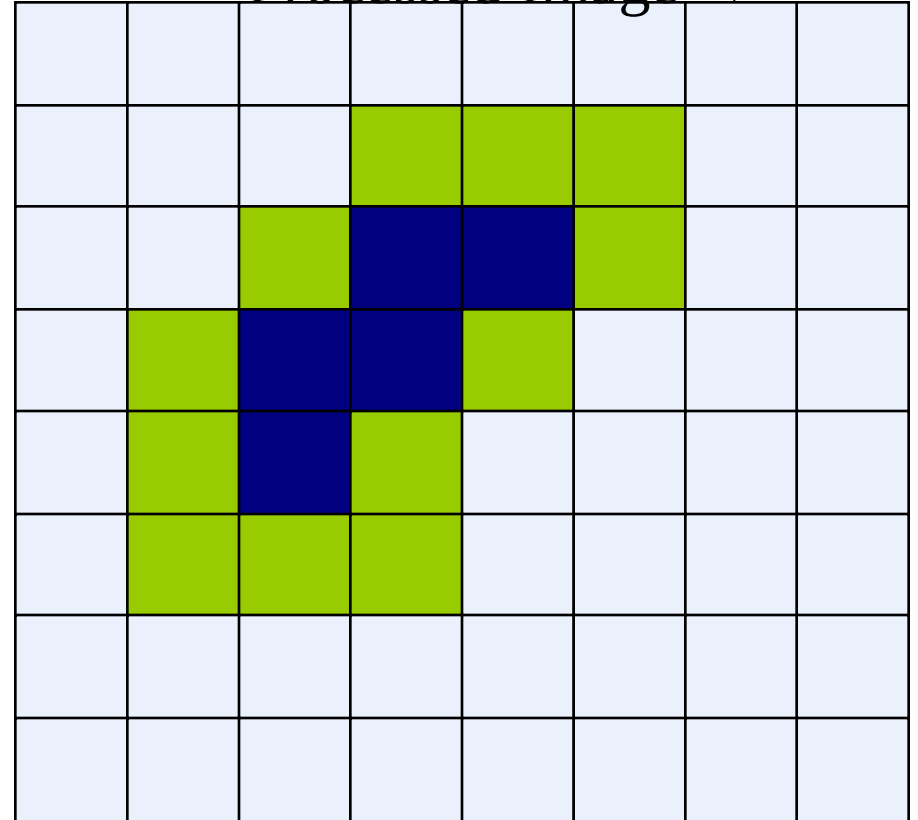
Erosion Example 2 cont.



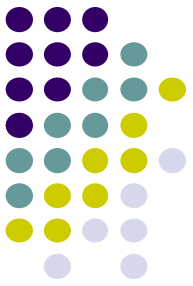
Original Image



Processed Image



Structuring Element



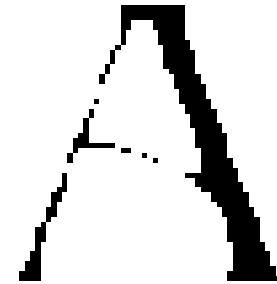
Erosion Example 3



Original image



Erosion by 3*3
square structuring
element

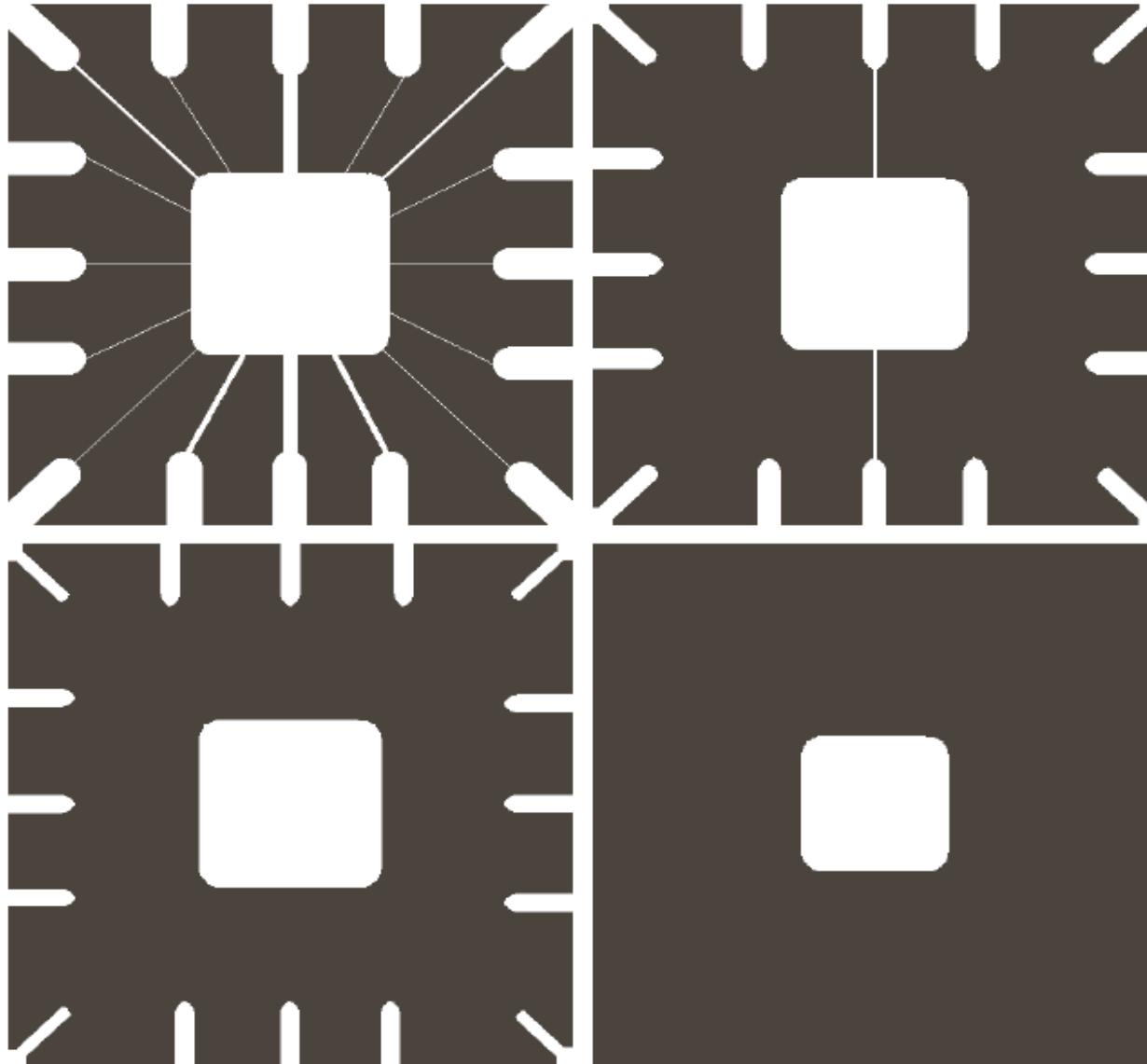


Erosion by 5*5
square structuring
element

Watch out: In these examples a 1 refers to a black pixel!



Erosion Example 4



a	b
c	d

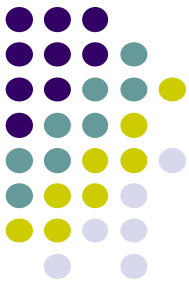
FIGURE 9.5 Using erosion to remove image components. (a) A 486×486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 , respectively. The elements of the SEs were all 1s.

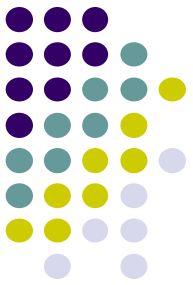




MATLAB CODE; MorphExample02

```
% read image
I=imread('cameraman.tif');
% convert to binary image
B=im2bw(I,0.3);
% create kernels
K3=ones(3);
K5=ones(5);
K7=ones(7);
K9=ones(9);
% apply erosion
B3=imerode(B,K3);
B5=imerode(B,K5);
B7=imerode(B,K7);
B9=imerode(B,K9);
% display images
imshow(B);title('original image');
figure, imshow(B3);title('eroded image with 3 by 3');
figure, imshow(B5);title('eroded image with 5 by 5');
figure, imshow(B7);title('eroded image with 7 by 7');
figure, imshow(B9);title('eroded image with 9 by 9');
```

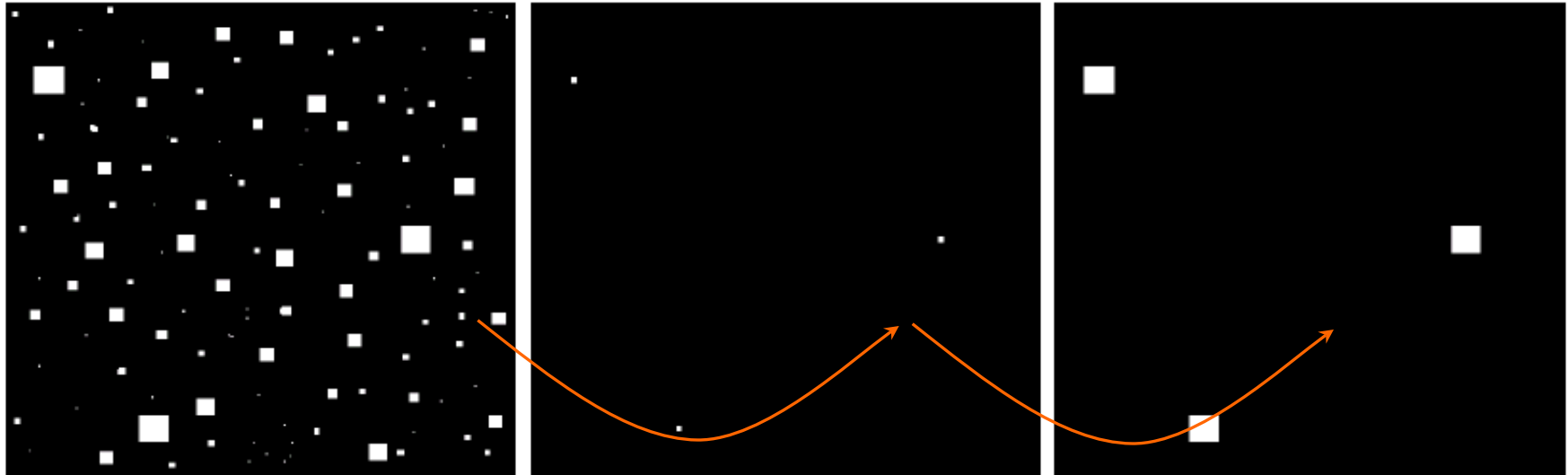
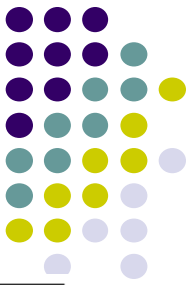




Dilation-Erosion Duality:

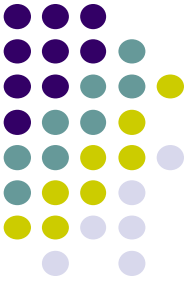
$$\begin{aligned}(A \ominus B)^c &= \{z \mid (B)_z \subseteq A\}^c = \{z \mid (B)_z \cap A^c = \emptyset\}^c \\ &= \{z \mid (B)_z \cap A^c \neq \emptyset\} = A^c \oplus \hat{B}\end{aligned}$$

Remember: $A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$



a b c

FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.



Thank you

for your attention